

Carbon Fibre Landing Gear – SimScale Results

Carbon Fibre Properties:

3k Twill Carbon Fiber

<https://www.easycomposites.co.uk/3d-print-to-carbon-fibre-development-kit>

https://www.fibreglast.com/product/3K_2_x_2_Twill_Weave_Carbon_Fiber_Fabric_01069

Young's Modulus: 231.66 – 240.63 GPa

Ultimate Tensile Strength: 3.45 GPa min 4.2 – 4.41 GPa

Warp	3K, 12-14
Filling	3K, 12-14
Strength	610-640 KSI
Tensile Modulus	33.6-34.9 MSI
Weave	2x2 Twill
Full Roll Length	100 yards
Color	Black
Weight	5.7 - 5.9 oz/sq yd

Regular Carbon Fiber:

▼ Mechanical

Young's Modulus	64.000 GPa	▲ ▼
Poisson's Ratio	0.10	▲ ▼
Shear Modulus	5000.000 MPa	▲ ▼
Density	1.600 g/cm ³	▲ ▼
Damping Coefficient	0.00	▲ ▼

▼ Strength

Yield Strength	380.000 MPa	▲ ▼
Tensile Strength	650.000 MPa	▲ ▼

☐ Thermally Treated

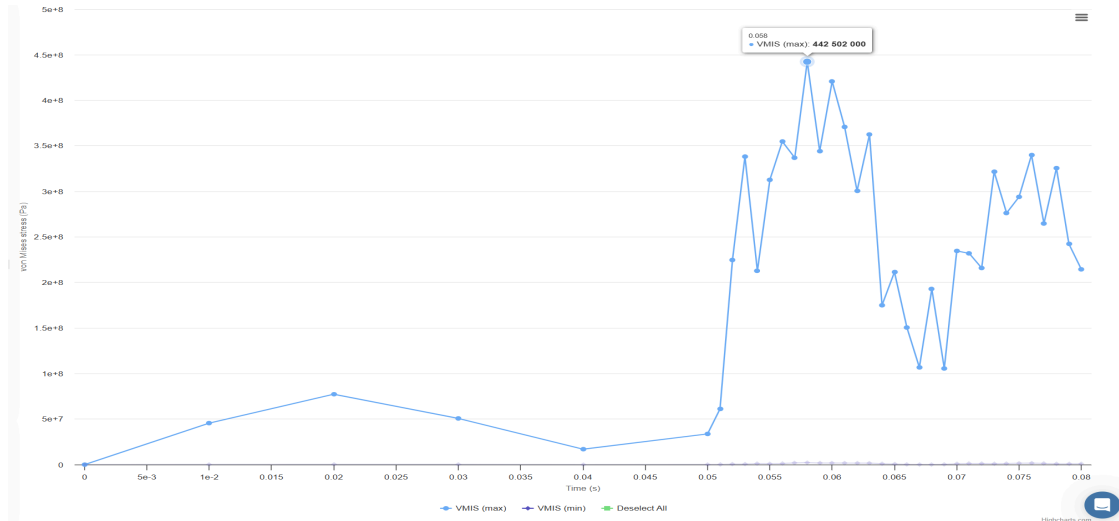
Purpose:

This paper compares the results of simulations run on SimScale for the landing gear planned to be used on the UCD UAV 25 with carbon fiber of two different types. Different landing scenarios are simulated, including ideal angle for the landing gear at different speeds, sideways tilted and backward tilted angles. Under each result, the maximum stress experienced in the landing gear is compared to the maximum tensile strength of the carbon fiber to determine whether it would fracture. An animation of each simulation is included with the results.

Ideal Angle Simulations

-1m/s

Regular Carbon Fiber:



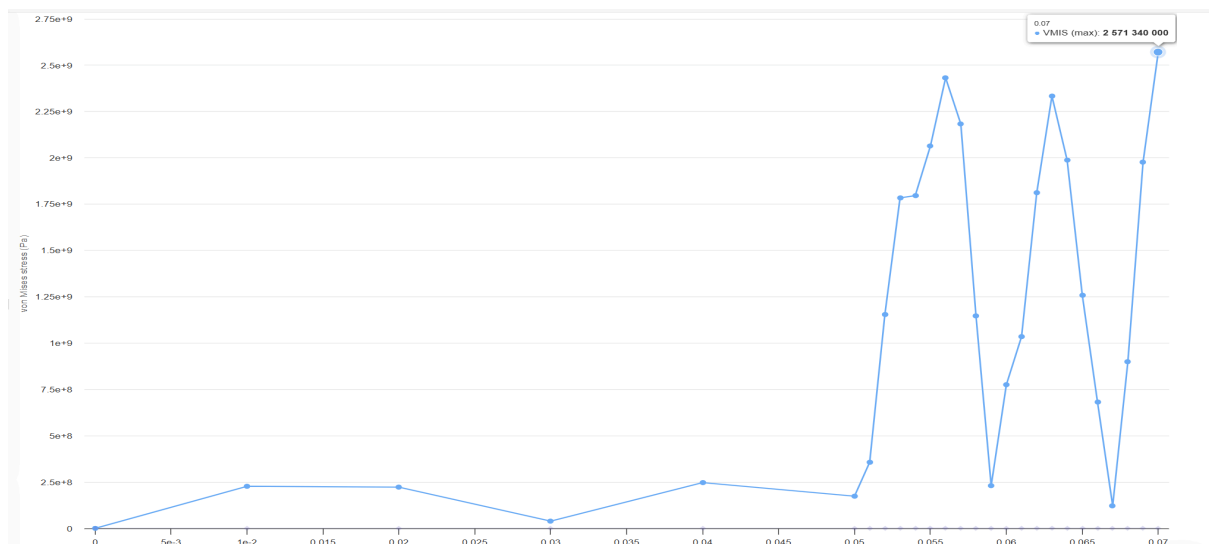
Max VMIS = 442.5 MPa

Material Max = 650 MPa



20241030-1620-17.6
147794.mp4

3K Twill Carbon Fiber



Max VMIS = 2.57 GPa

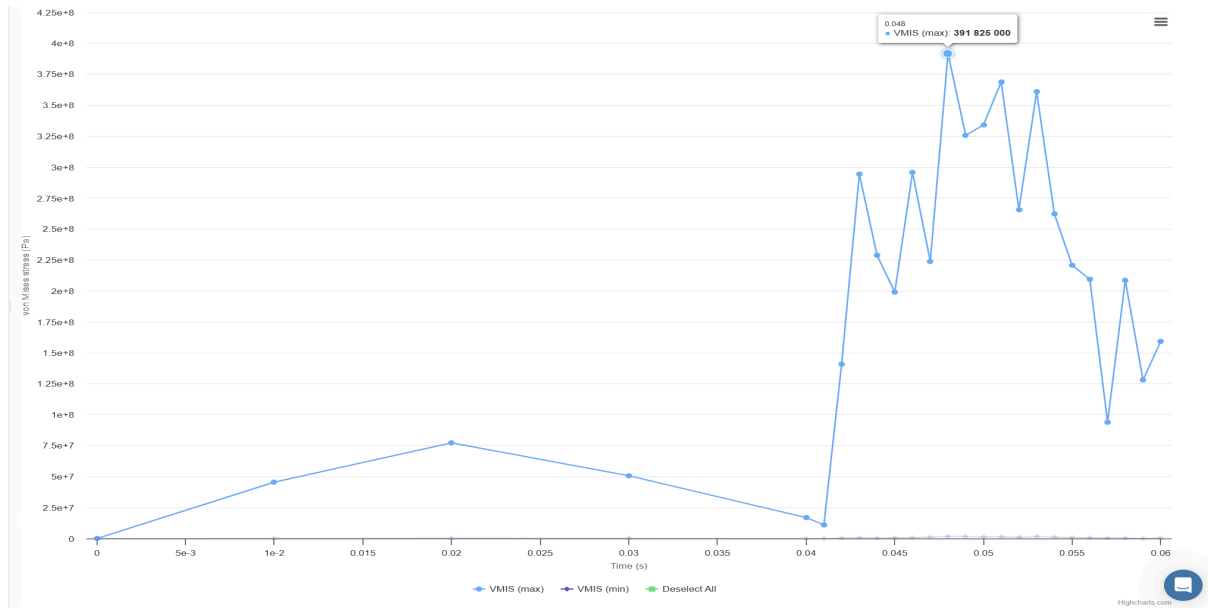
Material Max = 3.45 GPa



20241030-1616-52.1
046822.mp4

-3m/s

Regular Carbon Fiber:



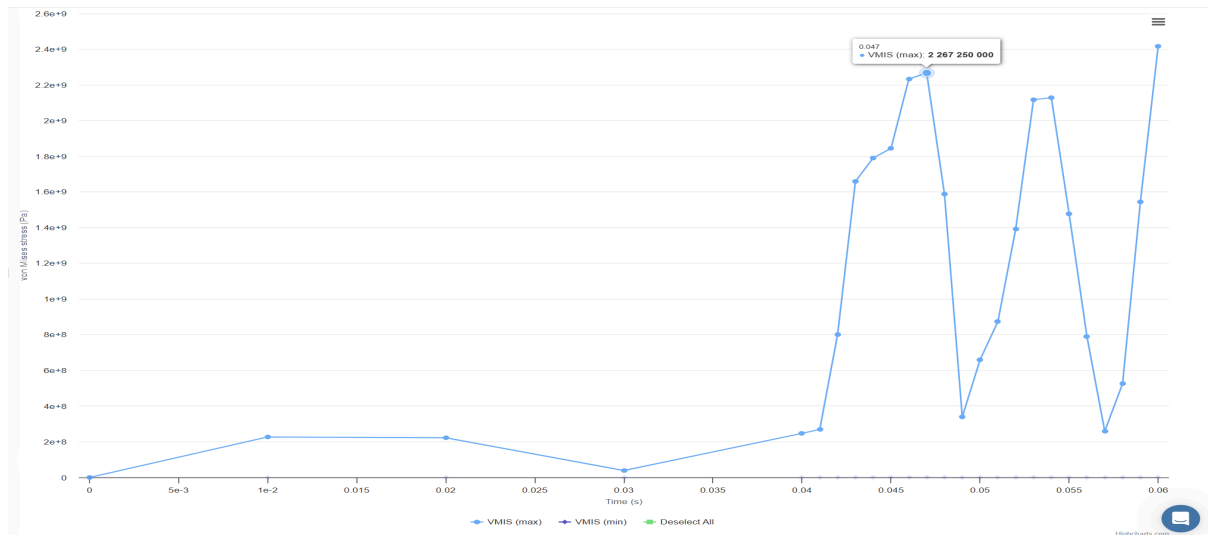
Max VMIS = 391.83 MPa

Material Max = 650 MPa



20241030-1133-06.5
346837.mp4

3K Twill Carbon Fiber:



Max VMIS during impact = 2.27 GPa

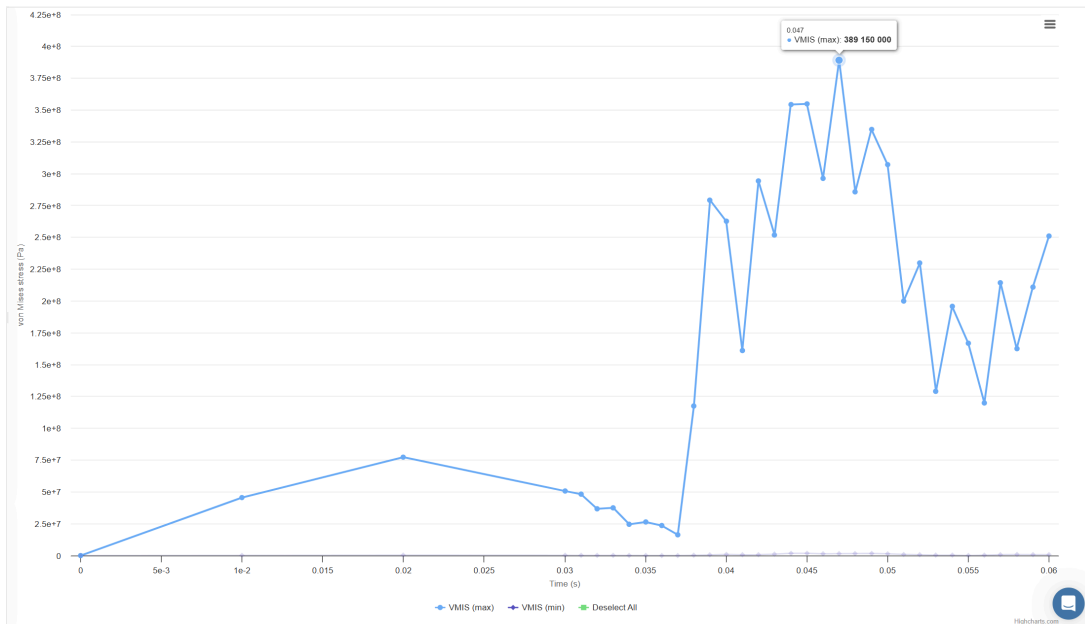
Material max = 3.45 GPa



20241030-1129-30.9
564695.mp4

-4m/s

Regular Carbon Fiber:



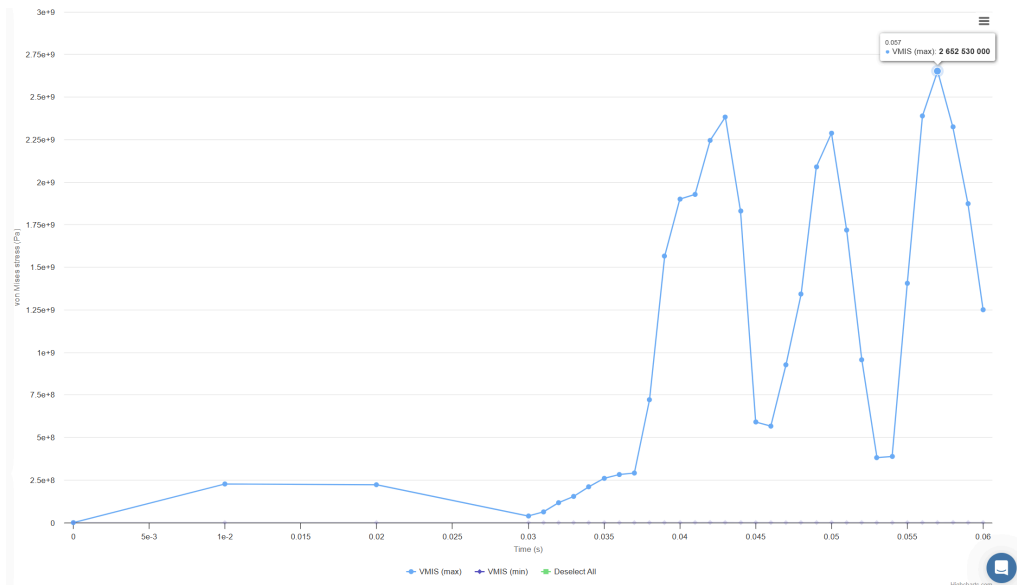
Max VMIS = 389.15 MPa

Material Max = 650 MPa



20241104-0943-11.9
655376.mp4

3K Twill Carbon Fiber:



Max VMIS = 2.652 GPa

Material Max = 3.45 GPa

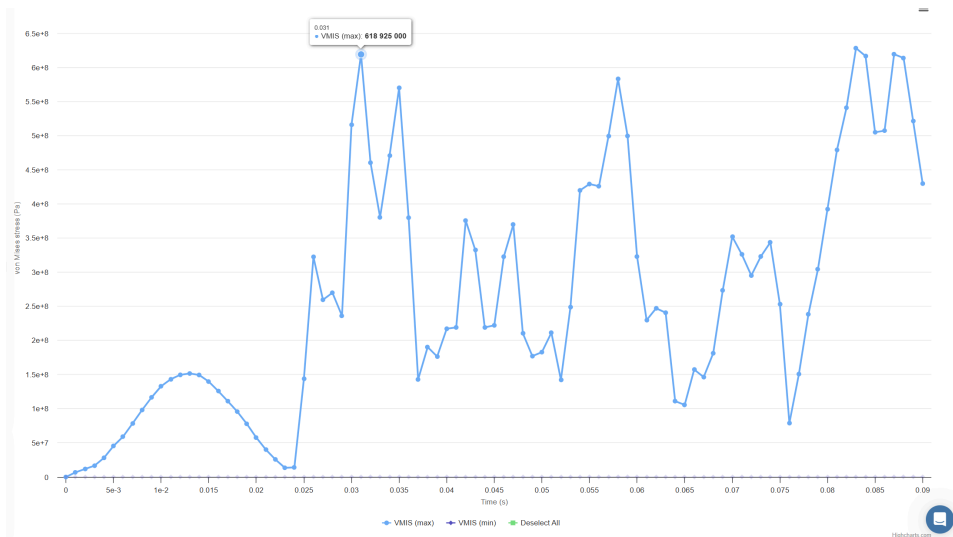


20241104-0946-53.1
971819.mp4

Tilted Landing Gear Simulations

10 Degree Backward Tilt

-1m/s

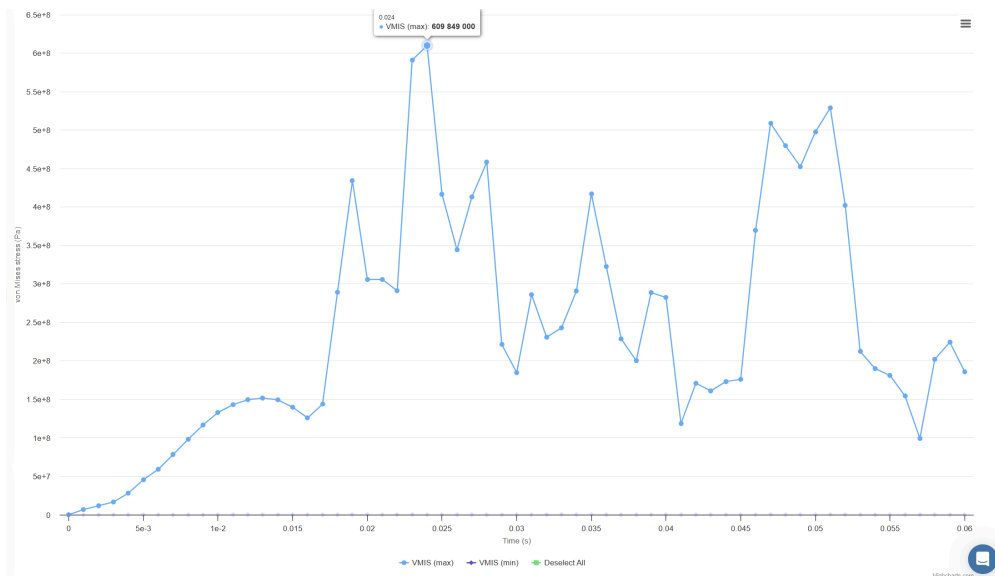


Max Von Mises = 618.925 MPa

Material Max = 650 MPa



20241106-1320-38.0
495447.mp4



Max Von Mises = 609.849 MPa

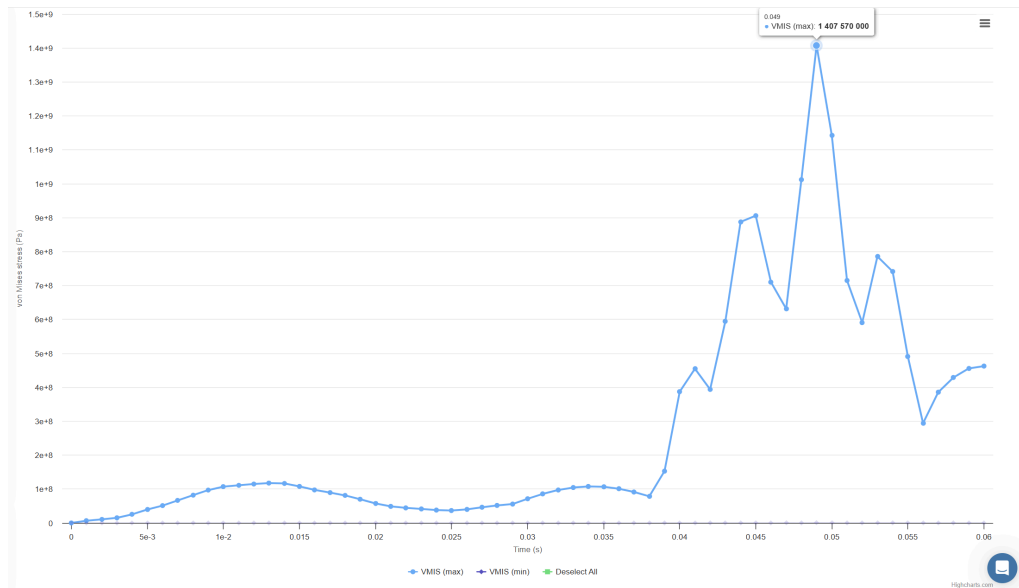
Material Max = 650 MPa



20241106-1324-00.9
568551.mp4

5 Degree Sideway Tilt

-3m/s



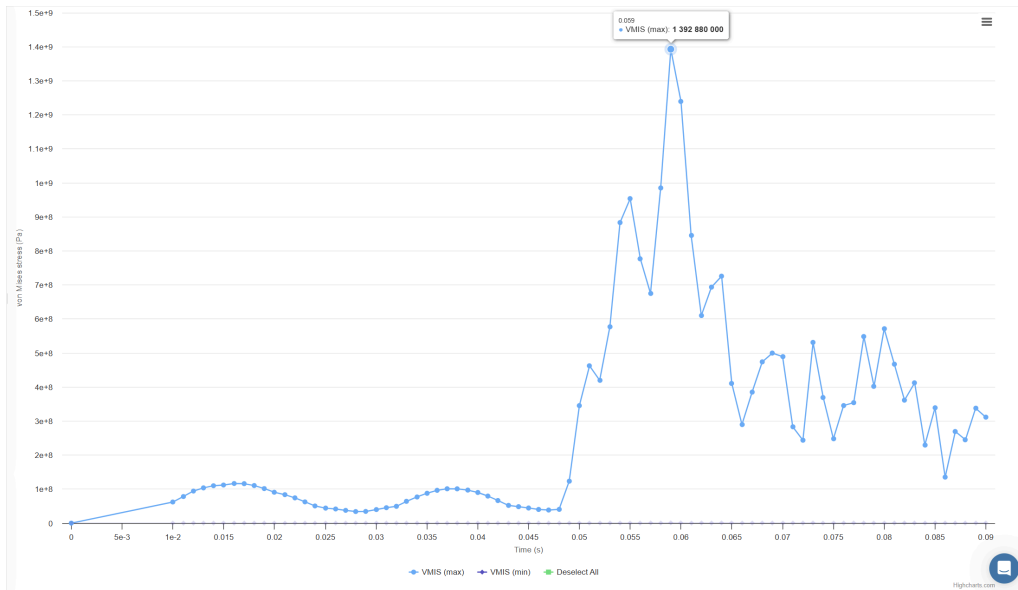
Max VMIS = 1.407 GPa

Material Max = 650 MPa



20241105-1725-00.6
666153.mp4

-1m/s



Max VMIS = 1.393 GPa

Material Max = 650 MPa



20241106-1326-38.6
294730.mp4